FORENSIC INTELLIGENCE AUDIT

Scenario A: Rating Trap & Defect Investigation

A Step-by-Step Practical Guide to What We Built, How We Analyzed It, and What We Found

Step 1: Problem Definition & Catalog Baseline

What is Scenario A? It represents *Rating Traps*: products showing high public star ratings (⭐ 4.0 / 5.0 or higher), but concealing high volumes of negative customer reviews (🔴 ≥ 35% to 100% negative feedback). Buyers trust the 4+ star badge, but end up receiving broken or defective merchandise.

ANALYZED REVIEWS	MATCHED PRODUCTS	FLAGGED TRAP SKUS	DECAY PEAK (2022)
50,000	**15,641**	**440**	**75.8%**
Amazon Appliances Batch	Unique Product SKUs	Scenario A Items (2.81%)	Negative Share Record

Step 2: Dual-Tier NLP & Root-Cause Breakdown

How we analyzed it: Instead of running slow models on all 50,000 reviews, we ran a Tier-1 fast rule filter to categorize text into positive, negative, and neutral sentiment. Then, on the 440 flagged trap products, we classified their negative reviews into exact root causes to check if delivery or product quality was to blame:

Failure Category	Complaints	Share (%)	What Customers Stated
Core Quality & Defect	243	48.89%	Motor stopped, cracked plastic, burned elements
Size & Compatibility	117	23.54%	Lid does not fit, wrong roller dimension, loose stud
General Dissatisfaction	103	20.72%	Cheap materials compared to original OEM part
Delivery & Shipping	26	5.23%	Late arrival, lost tracking, courier damage
Packaging Issues	8	1.61%	Crushed box, unsealed plastic bag

Key Insight: Over **72.4%** of issues stem directly from manufacturing quality and dimensional sizing flaws, while shipping is responsible for only **5.2%**.

Step 3: Buyer Authenticity & Quality Decay Over Time

Did competitors leave fake reviews? No. Our audit revealed that **694 negative reviews came from Verified Buyers** (67.3% complaint density), proving these are authentic paying customers experiencing real product failures.

When did quality collapse? We analyzed review dates from 2015 to 2023. While complaints hovered around 63% in earlier years and dipped to 55.7% in 2018, they escalated rapidly to 63.5% (2019), 69.5% (2020), and reached an all-time peak of **75.8% in 2022**. This proves older good reviews artificially hold up the star average while newer batches suffer heavy defect rates.

Step 4: Social Validation (Upvotes) & Automated Action Notices

Helpful Upvotes: When customers see a critical review, they click "Helpful". For product B07JJQR94P (4.0 Stars), a single negative warning collected **74 helpful upvotes** from fellow shoppers. By weighting complaints with upvotes, we ranked the most damaging trap products across the marketplace.

> **Step 5: Automated Vendor Enforcement Notices**
> *We built an automated generator that compiles the exact defect driver, failure percentages, and verified customer quotes into an official `VENDOR_NOTICE_[ASIN].txt` memo. This informs the supplier of their listing discrepancy and requires a 14-day corrective action plan before algorithmic de-ranking occurs.*